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MILITARY SPECIFICATION

CRANE, OVERHEAD TRAVELING, ELECTRIC, 5-TON CAPACITY; AND RUNWAY CONDUCTOR SYSTEM COMPONENTS

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a 5-ton capacity, electrically powered, overhead traveling crane with pushbutton pendant control, bridge and runway conductor system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-60 - Tape; Packing, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-104	-	Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-V-173	-	Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-L-2105	-	Lubricating Oil, Gear, Multipurpose
MIL-C-3774	-	Crates, Wood; Open, 12,000- and 16,000-Pound Capacity
MIL-E-16298	-	Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw Threads for the Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION, INC. (AISC)

Manual of Steel Construction.

(Application for copies should be addressed to the American Institute of Steel Construction, Inc., 101 Park Avenue, New York, New York 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

B30.16	Overhead Hoists (Underhung)
B30.17	Overhead and Gantry Cranes (Top Running Bridge, Single Girder, Underhung Hoist)

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Structural Steel
A275 - Magnetic Particle Examination of Steel Forgings
A700 - Packaging, Marking, and Loading Methods for Steel Products for Domestic Shipment

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B63 - Resistivity of Metallically Conducting Resistance and Contact Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CRANE MANUFACTURERS ASSOCIATION OF AMERICA, INC. (CMAA)

No. 70 - Specifications for Electric Overhead Traveling Cranes

(Application for copies should be addressed to the Crane Manufacturers Association of America, Inc., 1326 Freeport Road, Pittsburgh, Pennsylvania 15238.)

HOIST MANUFACTURER'S INSTITUTE (HMI)

HMI-100 - Electric Wire Rope Hoists

(Application for copies should be addressed to the Crane Manufacturers Association of America, Inc., 1326 Freeport Road, Pittsburgh, Pennsylvania 15238.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - Industrial Controls and Systems
MG1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification covers a 5-ton, overhead traveling crane which consists of load hook, hoist, traveling bridge, end trucks, wiring, pendant pushbutton control, bridge conductor system, trolley bumpers

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(stops), limit switches for bridge and hoist, runway-current collector shoes on one end truck, and runway conductor system. The span, lift, headroom, hook approaches, and operating clearances shall be as specified (see 6.2). The crane shall be rated in consonance with CMAA Standard No. 70 for class B service.

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete crane assembly, bridge and runway conductor system for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The crane assembly, bridge and runway conductor system shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the crane being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Components.

3.5.1 Overhead traveling crane. The crane component shall be a complete integral unit with end trucks spaced to operate in either position on a runway. Lubrication points and maintenance service areas shall be conveniently accessible. All warning labels and instructions attached to the crane and all safety tests performed after installation of this equipment shall be in accordance with ASME B30.16 and B30.17. The field wiring and complete electrification system connecting the runway current collectors with the bridge trolley and hoist motors shall be provided and installed. Screw threads shall conform to FED-STD-H28.

3.5.2 Bridge and end trucks. The bridge shall consist essentially of a single girder beam, stabilizing framework, drive shaft guides, trolley electrical conductor system supports, and overrunning end trucks alined to be operated on runway rails having a span of 34 feet, 2 1/2-inches center-to-center of the rails. The bridge beam shall conform to AISC Manual of Steel Construction, 18- by 7.50-inch-wide flange beam, or 18- by 6-inch American standard beam. The upper surface of the bridge beam lower flanges shall be finished in accordance with commercial practice. Bumper stops shall be installed at each end of the bridge and shall be located to engage the trolley bumpers at the limits of

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trolley travel. End trucks shall be either of the two-wheel central equalizer support design or of the outrigger design. Hook end approach distances shall not be greater than 4 feet 6 inches and 5 feet 0 inch from the respective runway ends. End trucks shall have a wheelbase of not less than 4 feet 11 inches ±1/2-inch. Wheels shall be ball or roller bearing mounted, double-flanged type with threads contoured to operate on American Society of Civil Engineers crane rails (30 pounds per yard). Drive wheels on opposing runway-track beams shall be connected with splined drive shafts supported from the bridge structure by supports equipped with ball or roller bearings. Structural steel used shall be in accordance with ASTM A36.

3.5.3 Bumpers and stops. Trolley bumpers shall be designed to absorb the travel energy (in the event of limit switch failure) of the trolley and hoist traveling at 50 percent of the maximum rated trolley speed. Spring-type trolley bumpers shall be provided on the trolley for both directions of travel. Bridge bumpers shall be designed to absorb the travel energy of the assembled bridge and hoist traveling at 50 percent of maximum rated speed. Spring-type bumpers shall be provided on each end truck for both directions of travel. Four track- or rail-mounted stops shall be provided for use on the runway.

3.5.4 Hoist and trolley. The hoist and trolley shall be a weatherproof, 2-speed, electric motor, 5-ton capacity cable hoist in conformance with HMI-100. Trolley wheels shall be flanged and shall have steel treads contoured to operate on the lower-track beam flanges. The hoist shall be suspended from the trolley with its long, horizontal dimension perpendicular to the track beam to provide for maximum trolley travel and minimum end approach clearance. Rope drums shall be right- and left-hand grooved for true vertical lift. The pendant pushbutton control station shall be suspended and shall control vertical hook travel, trolley travel, and bridge travel. Upper and lower hook travel limit switches shall be installed and set to prevent overtravel of the hook between upper and lower limits. The load hook shall be a heat treated, drop-forged steel, hoist hook with a swivel and safety latch. Hook nuts shall be keyed to hook shanks by means of a set screw or other similar, easily removable securing device. When inspection is in accordance with 4.3.1 there shall be no linear indication revealed by magnetic-particles. The hook shall be capable of supporting a 6.25-ton proof load, rotating through 360°, without damage. Rotation of the loaded hook shall not result in malfunction of rope leads.

3.6 Electrical.

3.6.1 Hoist and bridge. The crane assembly shall be designed for operation on 208 volt, 3 phase, 60 hertz electrical power. Motors, motor brakes, controllers, resistors, protection, wiring, and current collectors shall be in accordance with CMAA No. 70, as applicable. Upper and lower limit switches shall be provided in accordance with NEMA Standard ICS, part 2-225, as applicable to the duty class of the hoist. If the pendant control casing is metal, it shall be suitably grounded. The bridge shall be provided with a complete, enclosed safety type conductor system interconnecting the runway current collectors with the pendant control, the hoist, and the bridge end truck motors. All electrical parts and controls shall be weatherproof and grounded, in accordance with Article 610, NFPA No. 70.

3.6.2 Motors. Motors shall be 2-speed and rated in accordance with the following:

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Motion	Minimum hp	Speeds (fpm)
Hoist (hook lift)	7.50	17 (slow)
Trolley travel	0.750	75 (nominal maximum)
Bridge travel	3	175 (nominal maximum)

3.6.3 Controls. The strain lead shall support the pushbutton control below the trolley. A single pendant pushbutton control station shall be provided with the top control button at an elevation four feet above the floor line. Pushbuttons shall be clearly identified and arranged in the following order from top to bottom: BRIDGE FORWARD, BRIDGE REVERSE; TROLLEY FORWARD, TROLLEY REVERSE; UP, DOWN; RESET, STOP. Controls shall be for 2-speed motors. Initial adjustments of the limit switches shall bring the trolley or bridge travel to a stop 4 inches before striking the bumper (stop) at either end of the travel limit. The hoist limit switches shall be set to stop the hook 1 inch from the top position and allow a minimum lift of 16 feet. All electrically controlled brakes shall be actuated automatically by an interruption of the power supply. Interlocks shall be provided to prevent activation of more than one control at a time and shall direct reversing of any motion without first braking and bringing the previous motion to a stop. A "power on" indicating light shall be provided on the pendant pushbutton control case.

3.6.4 Bridge and runway conductor systems. Bridge and runway conductor system shall be the enclosed safety type. Conductors shall be rigid shapes of galvanized steel, copper-clad steel, or hard drawn copper proportioned to limit the total voltage drop in the conductors with full load current on the individual motors, to not more than 3 percent of the supply voltage. Supports shall be spaced in accordance with the manufacturer's recommendations. All supports, brackets, insulators, fastening devices, conductors, and power takeoffs necessary for complete installation shall be furnished. Conductor covers shall completely cover all live conductors and be shaped to prevent accidental contact of live conductors. Collectors shall be double-wheel or twin sliding-shoe type and shall match their respective electrification systems. Collector arms shall be insulated and spring-loaded and shall permit sufficient lateral and vertical movement to allow for imperfect track and for misalinement of the collector support in relation to the conductors. Runway electrification system components to enable constructing runway extensions, if specified (see 6.2), must be designed to match with runway electrification components furnished in accordance with this specification.

3.7 Performance. The assembled crane, with a rated hook load of 5 tons, shall be capable of not less than 2 hours continuous operation through a systematic cycle of timed high- and low-speed lifting starts, trolley travel starts, bridge travel starts, motion reversals, controlled stops, power interruption stops, and limit switch controlled stops without overheating, breakdown, or damage. This continuous cycle operation shall be at an average rate of not less than 100 (total) motion changes per hour and shall include not less than 5 full range lifts at maximum speed with 50 percent of the rated load on the hook.

3.7.1 Power supply. In ambient temperatures between 86°Fahrenheit (F) and 104°F, no electrical part shall increase in temperature beyond the limits allowed for corresponding parts in accordance with NEMA Standards MG1 and ICS, as applicable for class B duty hoist and crane service. The hoist shall attain steady lifting speed with rated load in not less than 1 foot nor more than 3 feet for either high or low speeds. Pushing the stop button shall immediately

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cease and hold any and all crane motions and shall override any faulty, accidentally applied, or stuck control. Interruption of electrical power supply shall return all controls to the off position. No control shall be effective unless full power is first connected. The load shall not drift, slip, nor drop when control buttons are operated, or when electrical power supply has been interrupted. Automatic, mechanical, and electrical load brakes shall support a 6.25-ton load without assistance from the motor or power supply for a minimum period of 4 hours (see 4.4).

3.8 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.9 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment (1,000 pounds-force per square inch (psi) or higher) will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.10 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.10.1 Stenciling. When specified (see 6.2), the load hook block, excluding hook, and pendant control station shall be painted yellow gloss. The maximum safe operating load capacity of the crane shall be permanently stenciled on both sides of the bridge girder, on the load block, and on the trolley in lettering in size to be readable from the floor. Bridge and Trolley direction shall be stenciled in accordance with control pushbutton operation, (see 3.6.3). Painting and stenciling shall be in accordance with manufacturer's current processes.

3.11 Instruction plates. The crane assembly shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall

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be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 0.125-inch diameter.

3.12 Identification marking. Identification shall be permanently and legibly marked directly on the crane assembly or on a corrosion-resisting metal plate securely attached to the crane at the source of manufacture. Identification shall include the manufacturer's model, serial number, name, and trademark to be readily identifiable to the manufacturer.

3.13 Workmanship.

3.13.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.13.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.13.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.13.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.13.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on a complete crane assembly, bridge and runway conductor system when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5.

4.3 Examination. Each crane, bridge and runway conductor system shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.3.1 Magnetic particle examination. The magnetic particle examination of steel forgings shall be in accordance with ASTM A275. A defect is defined as a surface or subsurface defect as follows. Surface defects have sharp distinct, clear-cut, tightly adhering patterns, characterized by laminar defects, forging laps, folds, and flakes. Subsurface defects produce less distinct, or fuzzy patterns which are broad rather than sharp, these are characterized by stringers of non metallic inclusions, cracks in underbeads of welds, and forging bursts.

4.4 Tests. Components, parts, subassemblies, and materials not produced by the crane manufacturer shall be source identified and either certified by their manufacturer as conforming to the referenced documents or tested and accepted as applicable to the referenced documents. First article items shall be inspected or certified with copies of test records as having been tested and approved in accordance with 4.3. Production items shall be tested or so certified in accordance with 4.4.1 through 4.4.3 as applicable. All defects shall be corrected. Uncorrectable defects shall be cause for rejection.

4.4.1 Production item tests. Production items shall be functionally operated, fit tested, measured, and examined, as appropriate, and compared with identically tested and accepted items, as applicable to the type.

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4.4.2 Component tests. The overhead traveling crane shall be operated with rated load through the cycle of operation in accordance with 3.7 for a period not less than 2 hours. During the cycle of operation, all speeds shall be measured and recorded. Heat-rise determinations shall be recorded for later analysis to determine conformance to NEMA Standard MG1 and ICS limits. A 6.25-ton test load shall be placed on the hook and rotated through two complete revolutions in each direction to determine brake-holding power and thrust-bearing function. The hoist and trolley subassembly shall be tested in accordance with ASME B30.16 as applicable for 2-speed electric motor, 5-ton rated load capacity hoists.

4.4.3 Bridge and runway system component tests. The bridge and runway system components shall be electrically tested for resistivity in accordance with ASTM B63. The terminal lug shall be attached to a number 2 AWG conductor and a load of 115 amperes applied. Measurement of terminal lug temperature shall be made one-half hour after the temperature has reached a constant value to determine conformance to 3.7.1. Following the temperature rise test at 115 amperes, overload tests of 125 percent of ampacity for 2 hours and 150 percent of ampacity for 5 minutes shall be made. Temperature measurements shall be made at the conclusion of each period to determine conformance to 3.7.1.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Wire rope. Wire rope shall be preserved with a dry film lubricant.

5.1.1.6 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

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5.1.1.7 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.8 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.9 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housing filled to the operating level with type P-10 preservative, grade as applicable. Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with lubricant conforming to MIL-L-2105, grade as applicable. Gears not operating on lubricating oil (SAE 10, 30, or 50) or lubricant (SAE 80 or 90) shall be filled to the operating level with the approved lubricant required for operation. Gear housings containing any of the above shall be identified with a weatherproof tag to indicate: "The housing is filled to the operating level with lubricant. Do not drain until first required lubrication change." Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

5.1.1.10 Hoist and trolley. All openings in the hoist, except those for the wire rope at the bottom of the hoist, shall be sealed with tape conforming to PPP-T-60, type IV.

5.1.1.11 Electrical motors. Preserve level A by the alternate method of MIL-E-16298.

5.1.1.12 Consolidated packaging. Complete crane assemblies shall have small disassembled items and small components packaged in close-fitting boxes conforming to PPP-B-636, class weather-resistant or PPP-B-601, domestic type, when an open crate (MIL-C-3774) is used for packing. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes.

5.1.2 Commercial. The complete crane assembly shall be preserved and packaged in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Hoist, trolley, end trucks, and accessories. The hoist, trolley, end trucks, consolidated packaged components, and accessories for each crane assembly shall be packed in a crate conforming to MIL-C-104, type I, style A, class optional. The contents shall be anchored, blocked, and braced in accordance with the crate specification.

5.2.1.2 Bridge and bus bars. The bridge girders shall be shipped loose without packing. The bus bars shall be packed in accordance with the requirements of ASTM A700 for bars and alloy steel.

5.2.2 Level B. Each complete crane assembly shall be packed, as specified, for level A, except that the crates shall conform to MIL-C-3774, type 1, style A. The contents shall be anchored, blocked, braced and waterproofed with a shroud in accordance with the crate specification.

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5.2.3 Commercial. The complete crane assembly shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The 5-ton capacity, electric, overhead traveling crane assembly is intended for installation on overhead runway rails in repair, assembly, or maintenance shop buildings, and for runway extensions which may go outside of the building.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of document if other than as specified (see 2.1.1).
c. Span, lift, headroom, hook approaches, and operating clearances as specified (see 3.1).
d. When first article is required (see 3.2 and 6.4).
e. Type of runway electrification components required (see 3.6.4).
f. When fungus resistance is required (see 3.8).
g. When lubrication other than as specified is required (see 3.9).
h. Color of paint for finish coat required (see 3.10).
i. Hook block and pendant control station paint required (see 3.10.1).
j. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CRDL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required (see 3.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one crane assembly and the bridge and runway conductor systems. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

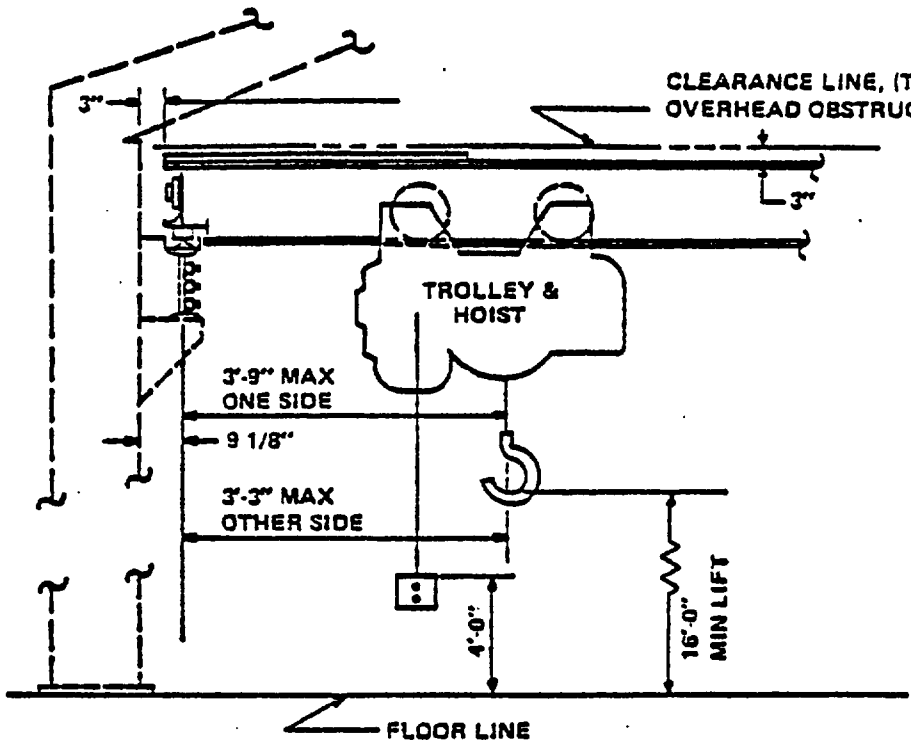
Navy - YD

Project No. 3950-0166

Review activity:

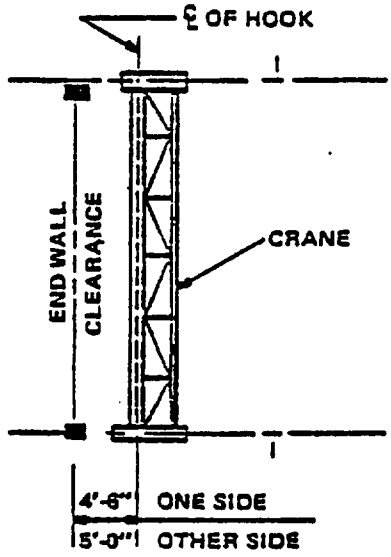
Air Force - 84

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CRANE DIMENSIONS
AND CLEARANCES

NO SCALE



PLAN OF CRANE

NO SCALE

Figure 1. Schematic arrangement clearance drawing.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side) Project No. 3950-0166

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-16536C	CRANE, OVERHEAD TRAVELING, ELECTRIC, 5-TON CAPACITY; AND RUNWAY CONDUCTOR SYSTEM COMPONENTS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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